DE BEERS GROUP

DIAMOND QUALITY, TRACEABILITY & PROCESS CONTROL

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Disclaimer: This is an independent educational case study created as part of my "100 Days Quality Management Challenge." De Beers Group is a separate organisation. This is not an official De Beers Group project, report, collaboration, or representation. The analysis applies general Quality Engineering principles using publicly available information.

QUALITY CHALLENGE

Diamond processing involves multiple stages where consistency, identification, grading and process control are important quality considerations.

Variation between processing stages or recurring quality issues can affect downstream quality, process efficiency and traceability.

QUALITY ENGINEERING QUESTION

How can process variation, quality risks and traceability-related issues be identified and controlled throughout the diamond value chain?

PROPOSED QUALITY APPROACH

TRACE → **MEASURE** → **MONITOR** → **ANALYSE**

- **TRACE:** Maintain identification & process history
- **MEASURE:** Collect critical quality data
- **MONITOR:** Detect abnormal variation
- **ANALYSE:** Apply Pareto, RCA & FMEA

CORRECT → **VERIFY** → **CONTROL**

- **CORRECT:** Address underlying causes
- **VERIFY:** Confirm corrective effectiveness
- **CONTROL:** Standardise & monitor continuously

CORE QUALITY TOOLS

FMEA
Identify and prioritise potential failure modes

Process Mapping
Understand critical processing stages

SPC & Trend Analysis
Monitor process stability and variation

Traceability Controls
Maintain product/process history

RCA
Identify underlying causes

Pareto Analysis
Prioritise significant quality problems

CAPA
Implement and verify corrective/preventive actions

EXPECTED QUALITY IMPACT

- ✓ Better process consistency
- ✓ Earlier detection of quality variation
- ✓ Stronger traceability
- ✓ Reduced recurring quality issues
- ✓ Better process predictability
- ✓ Improved quality-risk management
- ✓ Stronger prevention-based quality

QUALITY TAKEAWAY

"Quality is not only about inspecting the final product. It is about controlling variation, maintaining traceability and preventing quality risks throughout the process."

SUMMARY OF APPLIED QUALITY TOOLKIT

1. RISK & FLOW

FMEA & Process Mapping: Identifies high-risk processing stages and establishes standard input-output criteria for grading consistency.

2. MONITORING

SPC & Traceability: Utilises control charts to detect process drift early while locking digital assets to physical batch serials.

3. PROBLEM SOLVING

RCA & Pareto Analysis: Isolates the vital 20% of defect causes (e.g. cutting variance) using 5-Why and Fishbone frameworks.

4. ACTION & CONTROL

CAPA System: Drives standardized corrective actions to eliminate repeat defects and sustain continuous process control.